



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education
Advanced Subsidiary Level and Advanced Level

MATHEMATICS

9709/04

Paper 4 Mechanics 1 (M1)

October/November 2007

1 hour 15 minutes

Additional Materials: Answer Booklet/Paper
 Graph Paper
 List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

Where a numerical value for the acceleration due to gravity is needed, use 10 m s^{-2} .

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

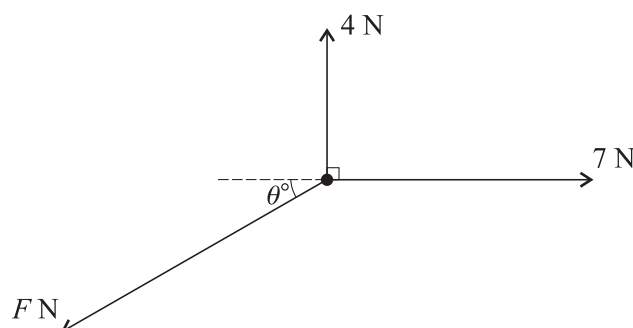
The total number of marks for this paper is 50.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of 4 printed pages.



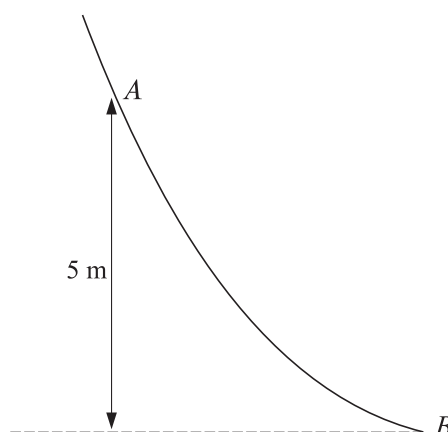
- 1 A car of mass 900 kg travels along a horizontal straight road with its engine working at a constant rate of P kW. The resistance to motion of the car is 550 N. Given that the acceleration of the car is 0.2 m s^{-2} at an instant when its speed is 30 m s^{-1} , find the value of P . [4]
- 2 A particle is projected vertically upwards from a point O with initial speed 12.5 m s^{-1} . At the same instant another particle is released from rest at a point 10 m vertically above O . Find the height above O at which the particles meet. [5]
- 3



A particle is in equilibrium on a smooth horizontal table when acted on by the three horizontal forces shown in the diagram.

- (i) Find the values of F and θ . [4]
- (ii) The force of magnitude 7 N is now removed. State the magnitude and direction of the resultant of the remaining two forces. [2]

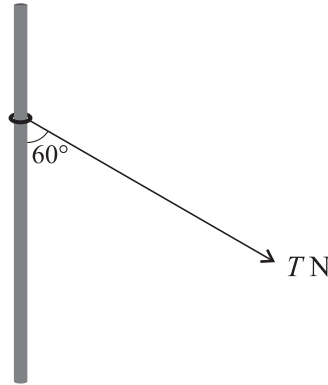
4



The diagram shows the vertical cross-section of a surface. A and B are two points on the cross-section, and A is 5 m higher than B . A particle of mass 0.35 kg passes through A with speed 7 m s^{-1} , moving on the surface towards B .

- (i) Assuming that there is no resistance to motion, find the speed with which the particle reaches B . [3]
- (ii) Assuming instead that there is a resistance to motion, and that the particle reaches B with speed 11 m s^{-1} , find the work done against this resistance as the particle moves from A to B . [3]

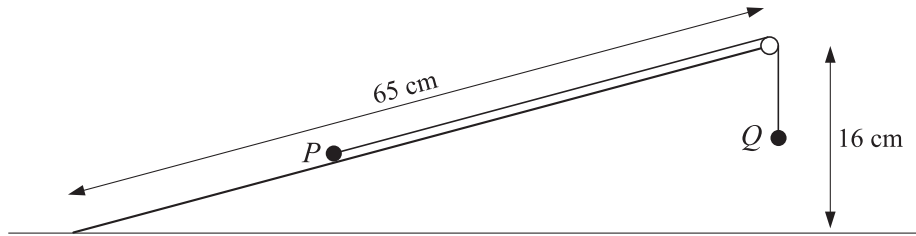
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A ring of mass 4 kg is threaded on a fixed rough vertical rod. A light string is attached to the ring, and is pulled with a force of magnitude $T\text{ N}$ acting at an angle of 60° to the downward vertical (see diagram). The ring is in equilibrium.

- (i) The normal and frictional components of the contact force exerted on the ring by the rod are $R\text{ N}$ and $F\text{ N}$ respectively. Find R and F in terms of T . [4]
- (ii) The coefficient of friction between the rod and the ring is 0.7 . Find the value of T for which the ring is about to slip. [3]

- 6
- (i) A man walks in a straight line from A to B with constant acceleration 0.004 m s^{-2} . His speed at A is 1.8 m s^{-1} and his speed at B is 2.2 m s^{-1} . Find the time taken for the man to walk from A to B , and find the distance AB . [3]
 - (ii) A woman cyclist leaves A at the same instant as the man. She starts from rest and travels in a straight line to B , reaching B at the same instant as the man. At time $t\text{ s}$ after leaving A the cyclist's speed is $k(200t - t^2)\text{ m s}^{-1}$, where k is a constant. Find
 - (a) the value of k , [4]
 - (b) the cyclist's speed at B . [1]
 - (iii) Sketch, using the same axes, the velocity-time graphs for the man's motion and the woman's motion from A to B . [3]



A rough inclined plane of length 65 cm is fixed with one end at a height of 16 cm above the other end. Particles P and Q , of masses 0.13 kg and 0.11 kg respectively, are attached to the ends of a light inextensible string which passes over a small smooth pulley at the top of the plane. Particle P is held at rest on the plane and particle Q hangs vertically below the pulley (see diagram). The system is released from rest and P starts to move up the plane.

- (i) Draw a diagram showing the forces acting on P during its motion up the plane. [1]
- (ii) Show that $T - F > 0.32$, where T N is the tension in the string and F N is the magnitude of the frictional force on P . [4]

The coefficient of friction between P and the plane is 0.6.

- (iii) Find the acceleration of P . [6]